

Industry Problem Statements 2026

10 real-world industry problems across 5 sectors

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■ Note to Participants

The problem statements below are provided as inspiration and serve as a starting point — they are not a definitive or exhaustive list. You are encouraged to bring your own problem statements, whether drawn from your industry experience or beyond the domains listed here. Innovation is not limited to what is on this page.

■■ Defense & Aerospace

D1 Mission Readiness & Predictive Maintenance Copilot

■ Critical Now

Military organisations cannot reliably determine whether aircraft, vehicles, and equipment are mission-ready. Maintenance runs on fixed calendar schedules regardless of actual component condition. HUMS (Health & Usage Monitoring System) sensor data that could predict failures weeks in advance sits unanalysed. When platforms fail unexpectedly, operational readiness drops and recovery takes weeks. The US military spends \$90B/year on maintenance — shifting to predictive approaches saves billions.

■ Your Challenge

Build a Bob Copilot that ingests sensor data and service records to identify non-ready assets, explain each readiness issue, predict which components will fail before the next mission window, and recommend a prioritised maintenance plan.

D2

Threat Intelligence Correlation & Alert Prioritisation Assistant

■ Critical Now

Defence analysts receive thousands of alerts daily from SIEM systems, satellite feeds, cyber sensors, and intelligence reports — all in different formats. No human team can read them all. Missing a genuine threat is catastrophic; chasing false positives wastes critical resources. Threat assessments must also be produced in structured BLUF (Bottom Line Up Front) format so commanders get a clear picture in minutes.

■ Your Challenge

Build a Bob solution that ingests multi-source threat feeds, correlates alerts to separate genuine threats from false positives, maps attacker techniques to the MITRE ATT&CK framework, and generates prioritised BLUF investigation summaries for commanders.

■ Semiconductor

S1

Wafer Yield Root Cause & Defect Pattern Analyser

■ Critical Now

At 3nm/5nm chip nodes, a 1% yield drop costs tens of millions per month. Root causes hide across thousands of equipment sensors, process parameters, and defect images. Engineers spend weeks finding the cause manually — every day of delay is lost revenue. Process engineers also need to predict which upcoming batches are at risk before they run, not after they fail.

■ Your Challenge

Build a Bob solution that analyses wafer lot data and defect reports to identify patterns, rank root causes by probability, recommend corrective actions, and flag upcoming batches whose process parameters historically correlate with low yield.

S2**Fab Bottleneck & Supply Chain Risk Advisor****■ Critical Now**

Chip lead times are 26–52 weeks. Bottlenecks at specific process steps (lithography, etch, CVD) cause cascading delivery delays — the 2021 chip shortage halted auto factories for months. Simultaneously, single-source suppliers for critical materials (gallium, neon, photoresists) create catastrophic risk: China controls 80%+ of gallium and germanium, both now under export controls.

■ Your Challenge

Build a Bob solution that identifies fab bottleneck tools from WIP queue data, predicts downstream delivery impact, and simultaneously evaluates the supply chain for single-point-of-failure suppliers and geopolitical concentration risk.

■ Pharma & Biotech**P1****Clinical Trial Risk Monitor & Protocol Deviation Detector****■ Critical Now**

A major clinical trial has 5,000+ patient visits across 200+ sites. Protocol deviations — missed visits, wrong dosing, banned co-medications — go undetected until the FDA audit. A single rejected submission delays drug approval by 6–12 months and costs \$50–100M. Risk managers need real-time visibility into which sites are highest risk before problems escalate.

■ Your Challenge

Build a Bob solution that compares patient records against the protocol specification, classifies each deviation by severity (ICH E6 GCP: major/minor/administrative), scores site-level risk using leading indicators, and generates CAPA-ready reports with recommended mitigations.

P2**Drug Safety Signal Detector & Regulatory Submission Readiness Checker****■ Critical Now**

FDA's FAERS database has 20M+ adverse event reports. Vioxx caused 27,000+ heart attacks before its signal was acted on. Separately, a drug approval CTD dossier spans 100,000+ pages across 5 modules — one missing section gets it rejected, costing 6–12 months and \$50–100M. Both problems share the same root cause: too much complex data for manual review.

■ Your Challenge

Build a Bob solution with two modes: (1) Signal Detection — cluster adverse event reports and calculate PRR statistics to flag emerging safety signals. (2) Submission Readiness — check a dossier outline against ICH M4 CTD requirements, score completeness per module, and generate a gap report.

■ Utilities — Power, Energy, Transmission & Distribution

U1

Power Outage Prediction & Grid Equipment Failure Advisor

■ Critical Now

Power transformer and substation failures cause blackouts costing utilities \$1M+/hour and affecting millions of people. Most utilities still use calendar-based maintenance, while sensors already measuring temperature, vibration, partial discharge, and oil quality show failure signatures weeks in advance. Weather events compound the risk — but sensor data and weather forecasts are never combined in time to act.

■ Your Challenge

Build a Bob solution that combines asset health sensor data, weather forecasts, and historical incident records to predict outage-prone areas and at-risk equipment, rank assets by grid impact severity, and generate a prioritised maintenance and crew pre-positioning plan.

U2

Grid Load Optimisation & Renewable Energy Performance Advisor

■ Critical Now

The US curtailed ~8 TWh of clean energy in 2023 — renewable electricity switched off because the grid couldn't absorb it. At the same time, unexpected demand spikes cause grid instability. Operators must forecast load spikes, balance renewable generation, and detect underperforming solar/wind assets simultaneously — with very limited real-time decision support today.

■ Your Challenge

Build a Bob solution that forecasts demand spikes, recommends load-balancing actions, detects anomalies in renewable energy performance, identifies root causes per underperforming asset, and generates an integrated operator optimisation brief with a curtailment minimisation plan.

■ Logistics & Ports

L1

Container Congestion Predictor & Port Operations Optimiser

■ Critical Now

The 2021 LA/Long Beach port backlog had 100+ ships waiting offshore for weeks, costing global supply chains \$10B+. Port operators allocate berths, cranes, and yard space across hundreds of vessels manually in spreadsheets. Congestion hotspots are identified reactively — after vessels are already queuing — and alternate routing decisions come too late to help.

■ Your Challenge

Build a Bob solution that predicts congestion hotspots using vessel schedules and berth capacity data, recommends alternate routing strategies, optimises berth and crane assignments, and generates a 72-hour port operations plan for shift supervisors.

L2

Supply Chain Disruption Assistant & Fleet Utilisation Optimizer

■ Ongoing Pain

Supply chain disruptions — weather events, port strikes, geopolitical crises — cascade across hundreds of active shipments in ways that are impossible to track manually. Fleet assets (trucks, containers, vessels) sit idle while other routes are overloaded. Cold chain shipments (vaccines, perishables) are especially vulnerable — a single temperature excursion across any leg can spoil a \$500K+ cargo, but breaches are only discovered at delivery when it is too late.

■ Your Challenge

Build a Bob solution that identifies which shipments are affected by an active disruption, recommends re-routing or carrier alternatives, identifies idle fleet assets for redeployment, and monitors cold chain IoT sensor logs to detect temperature excursions and classify their regulatory severity before delivery.

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